

In class Activity – Feb 08, 2023

LAB 03 : Create Directory Command and “chmod”

1. Login using your non-root user account.

\$ **login** : **van**

2. Go to your home directory and create a directory called **tmpinclass**.

\$ **mkdir tmpinclass** : create a directory in user **van**

3. Now, go the **tmpinclass** directory and create a directory tree with at least 3 directories and at least two files in each directory.

\$ **cd tmpinclass** : go to **tmpinclass** directory

\$ **mkdir EXCISE_01** : create a directory in **tmpinclass**

\$ **mkdir EXCISE_02** : create a directory in **tmpinclass**

\$ **mkdir EXCISE_03** : create a directory in **tmpinclass**

\$ **cd EXCISE_01** : go to **EXCISE_01** directory

\$ **touch excise_01_1** : create a file in **EXCISE_01**

\$ **touch excise_01_2** : create a file in **EXCISE_01**

\$ **touch excise_01_3** : create a file in **EXCISE_01**

\$ **cd ..** : go back **tmpinclass** directory

\$ **cd EXCISE_02** : go to **EXCISE_02** directory

\$ **touch excise_02_1** : create a file in **EXCISE_02**

\$ **touch excise_02_2** : create a file in **EXCISE_02**

\$ **touch excise_02_3** : create a file in **EXCISE_02**

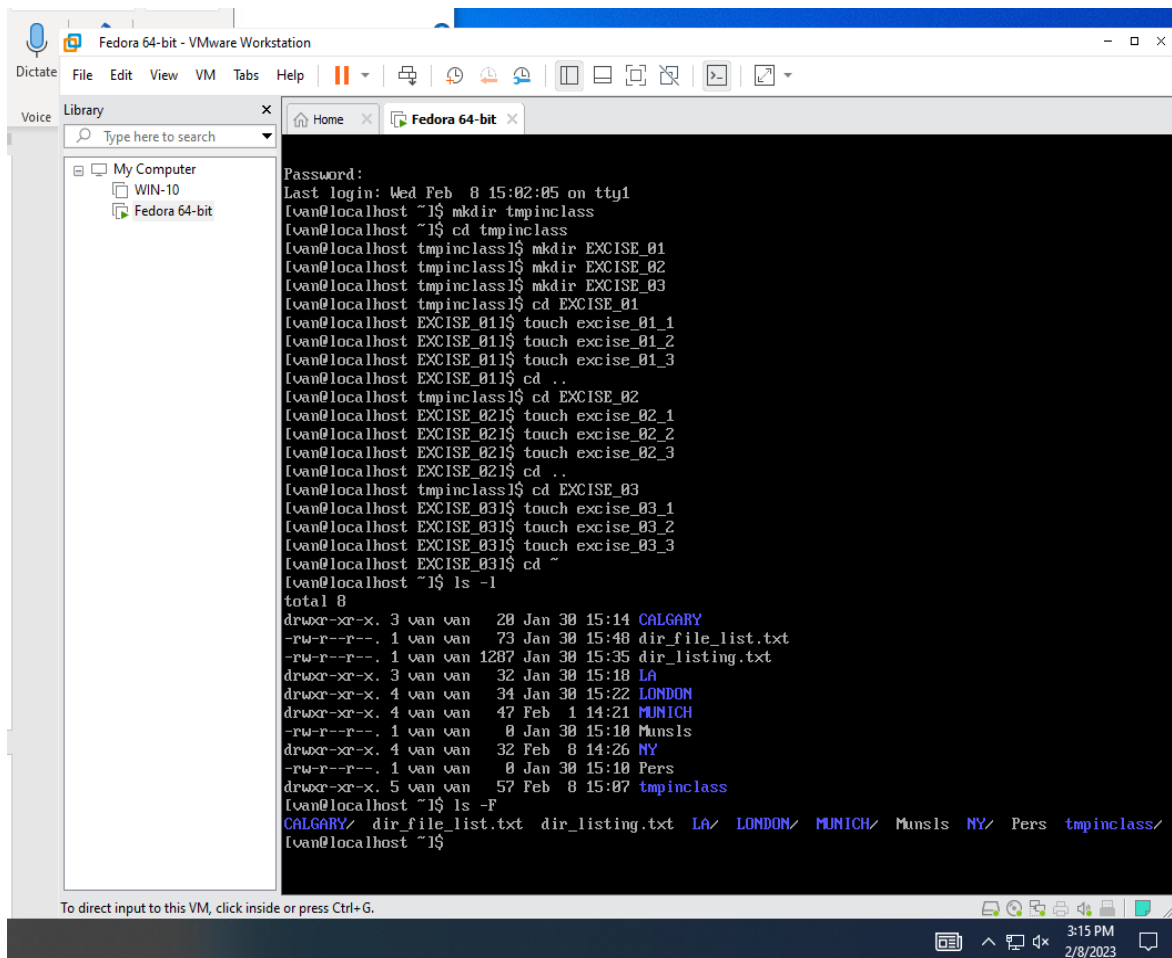
\$ **cd ..** : go back **tmpinclass** directory

\$ **cd EXCISE_03** : go to EXCISE_03 directory

\$ **touch excise_03_1** : create a file in EXCISE_03

\$ **touch excise_03_2** : create a file in EXCISE_03

\$ **touch excise_03_3** : create a file in EXCISE_03



The screenshot shows a VMware Workstation window titled "Fedora 64-bit - VMware Workstation". The interface includes a menu bar (File, Edit, View, VM, Tabs, Help), a toolbar, and a left sidebar with a "Library" pane showing "My Computer" with "WIN-10" and "Fedora 64-bit". The main terminal window displays the following commands and output:

```
Password:
Last login: Wed Feb  8 15:02:05 on tty1
[van@localhost ~]$ mkdir tmpinclass
[van@localhost ~]$ cd tmpinclass
[van@localhost tmpinclass]$ mkdir EXCISE_01
[van@localhost tmpinclass]$ mkdir EXCISE_02
[van@localhost tmpinclass]$ mkdir EXCISE_03
[van@localhost tmpinclass]$ cd EXCISE_01
[van@localhost EXCISE_01]$ touch excise_01_1
[van@localhost EXCISE_01]$ touch excise_01_2
[van@localhost EXCISE_01]$ touch excise_01_3
[van@localhost EXCISE_01]$ cd ..
[van@localhost tmpinclass]$ cd EXCISE_02
[van@localhost EXCISE_02]$ touch excise_02_1
[van@localhost EXCISE_02]$ touch excise_02_2
[van@localhost EXCISE_02]$ touch excise_02_3
[van@localhost EXCISE_02]$ cd ..
[van@localhost tmpinclass]$ cd EXCISE_03
[van@localhost EXCISE_03]$ touch excise_03_1
[van@localhost EXCISE_03]$ touch excise_03_2
[van@localhost EXCISE_03]$ touch excise_03_3
[van@localhost EXCISE_03]$ cd ~
[van@localhost ~]$ ls -l
total 8
drwxr-xr-x. 3 van van 20 Jan 30 15:14 CALGARY
-rw-r--r--. 1 van van 73 Jan 30 15:48 dir_file_list.txt
-rw-r--r--. 1 van van 1287 Jan 30 15:35 dir_listing.txt
drwxr-xr-x. 3 van van 32 Jan 30 15:18 LA
drwxr-xr-x. 4 van van 34 Jan 30 15:22 LONDON
drwxr-xr-x. 4 van van 47 Feb  1 14:21 MUNICH
-rw-r--r--. 1 van van  8 Jan 30 15:10 MunsIs
drwxr-xr-x. 4 van van 32 Feb  8 14:26 NY
-rw-r--r--. 1 van van  8 Jan 30 15:10 Pers
drwxr-xr-x. 5 van van 57 Feb  8 15:07 tmpinclass
[van@localhost ~]$ ls -F
CALGARY/ dir_file_list.txt dir_listing.txt LA/ LONDON/ MUNICH/ MunsIs NY/ Pers tmpinclass/
[van@localhost ~]$
```

At the bottom of the window, a status bar indicates "To direct input to this VM, click inside or press Ctrl+G." and the system clock shows "3:15 PM 2/8/2023".



4. Check if you have access to change the permissions of the directories (Use utility **sudo** if needed): First directory, second directory and third directory.

\$ **tree tmpinclass** : check the directories just created

\$ **ls -l** : Check the permissions of the directories

```

└─ Ng$3
  └─ Pers
    └─ tmpinclass
      └─ EXCISE_01
        ├── excise_01_1
        ├── excise_01_2
        └── excise_01_3
      └─ EXCISE_02
        ├── excise_02_1
        ├── excise_02_2
        └── excise_02_3
      └─ EXCISE_03
        ├── excise_03_1
        ├── excise_03_2
        └── excise_03_3

17 directories, 43 files
[van@localhost ~]$ ls -l
total 8
drwxr-xr-x. 3 van van 28 Jan 30 15:14 CALGARY
-rw-r--r--. 1 van van 73 Jan 30 15:48 dir_file_list.txt
-rw-r--r--. 1 van van 1287 Jan 30 15:35 dir_listing.txt
drwxr-xr-x. 3 van van 32 Jan 30 15:18 LA
drwxr-xr-x. 4 van van 34 Jan 30 15:22 LONDON
drwxr-xr-x. 4 van van 47 Feb 1 14:21 MUNICH
-rw-r--r--. 1 van van 0 Jan 30 15:10 MunsIs
drwxr-xr-x. 4 van van 32 Feb 8 14:26 NY
-rw-r--r--. 1 van van 0 Jan 30 15:10 Pers
drwxr-xr-x. 5 van van 57 Feb 8 15:07 tmpinclass
[van@localhost ~]$ ls -F
CALGARY/ dir_file_list.txt dir_listing.txt LA/ LONDON/ MUNICH/ MunsIs NY/ Pers tmpinclass/
[van@localhost ~]$ ls -a
. .bash_history .bash_profile CALGARY dir_listing.txt LONDON MunsIs Pers
. .bash_logout .bashrc dir_file_list.txt LA MUNICH NY tmpinclass
[van@localhost ~]$ ls -c
tmpinclass NY MUNICH MunsIs dir_file_list.txt dir_listing.txt LONDON LA CALGARY Pers
[van@localhost ~]$

```



5. Set the following:
6. First directory: owner should have read write and execute, group should have read and write, others should have read.

```
$ chmod u+rw EXCISE_01 :assign user van can read, write and execute
$ chmod g+rw EXCISE_01 : assign group can read and write
$ chmod o+r EXCISE_01: To give only read permission for others.
```

7. Second directory: owner should have read and write, group should have read, others should have read.

```
$ chmod u+rw EXCISE_02 : assign user van can read, write and execute
$ chmod g+r EXCISE_02 : assign group can read
$ chmod o+r EXCISE_02 : To give only read permission for others.
```

8. Third directory: all should have read permission

```
$ chmod a=r EXCISE_02 : to give only read permission for everyone.
```

9. Enter the first directory and use symbolic way to change the permissions of the file. You can select what permission each file has. Every file should have different permission.

Changing Permissions in Symbolic Mode

```
$ chmod uo+rx EXCISE_01 :assign user and others can read, and execute
```

10. Enter the second directory and use symbolic way to change the permissions of the file. You can select what permission each file has. Every file should have same permission.

```
$ chmod go+r EXCISE_02 : assign group and others can read
```

11. Enter the third directory and use numeric way to change the permissions of the file. You can select what permission each file has. Every file should have same permission.

```
$ chmod 764 EXCISE_03 : user can read, write and execute, group can read and write and others can read.
```

Short Answers (Ignore SUID, SGID and sticky bits):

1. If we want to set **rw-rw-rw-** permissions for the directory, what mask should we set? Write the command.

```
# [ chmod 774 < file name> ] : user can read, write and execute, group can read, write and execute and others can read.
```

2. If we want to set **rw-rw-r** permissions for the directory, what mask should we set? Write the command.

```
# [ chmod 664 < file name> ] : user can read and write, group read and write, others can read.
```

3. If the umask is set to 033 for a directory, what permissions does it have?

[**chmod** - - - wx - wx < file name>] : **group** write and execute ,
others write and execute.

4. If a umask is set to 022 for a file, what permissions does it have?

[**chmod** - - - w - -w - < file name>] : **group** write, **others** write

